

Moving charged particle

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Radiation loss

$$\Delta E = \frac{2\pi}{\omega} P = \frac{2\pi\rho}{c\beta} P = \frac{4\pi}{3} \frac{q^2}{\rho} \beta^3 \gamma^4 \propto \frac{\gamma^4}{\rho} \quad (1)$$

1 period = 1 revolution $\gamma = \frac{E}{mc^2}$

Ask for corresponding energies at some ΔE for protons vs electrons.

$$m_p \approx 2000 m_e \quad (2)$$

$$\text{same } \Delta E_p = \Delta E_{el} \implies \gamma_p^4 = \gamma_{el}^4$$

$$\frac{E_p}{m_p c^2} = \frac{E_{el}}{m_{el} c^2} \quad (3)$$

LEP: $E_{el} \approx 50 \text{ GeV}$ and LHC $E_p \approx 100 \text{ TeV}$. LHC currently $E_p \approx 13 \text{ TeV}$.

Limiting factor not radiator loss but the magnetic field required to keep protons and their circular trajectory.

0.0.1 (Lorentz boost in x-direction of electrostatic field HW1,Q4)

$$\vec{E}(\vec{x}, t) = \frac{q\vec{x}}{|\vec{x}|^3}, \vec{B}(\vec{x}, t) = 0 \quad (4)$$

$$E'_x = E_x \quad (5)$$

$$E'_y = \gamma(E_y - \beta B_z) = \gamma E_y \quad (6)$$

$$E'_z = \gamma(E_z + \beta B_y) = \gamma E_z \quad (7)$$

$$B'_x = B_x = 0 \quad (8)$$

$$E'_y = \gamma(B_y + \beta E_z) = \gamma\beta E_z \quad (9)$$

$$E'_z = \gamma(B_z - \beta E_y) = -\gamma\beta E_y \quad (10)$$

Lorentz covariant combination

$$\vec{E} \cdot \vec{B}, \vec{E}^2 - \vec{B}^2 \quad (11)$$

These can be checked see hw

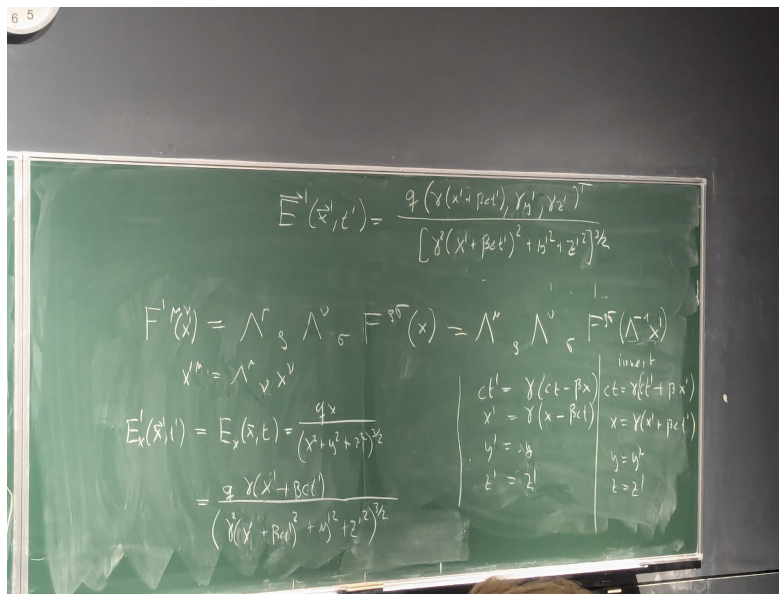


Figure 1: 2.1.jpg

$$E_{ret}(\vec{x}, t) = \left[\frac{q(\vec{n} \cdot \vec{\beta})}{R^2 \gamma^2 (1 - \beta \cdot \vec{n})^3} \right]_{ret} \quad (12)$$

which is a velocity field.

This should match the expression in the figure above.

$$\vec{\beta} = (-\beta, 0, 0)^T \quad (13)$$

$$\vec{r}(t) = (-\beta ct, 0, 0)^T \quad (14)$$

Trajectory of point charge.